

Multiple Choice (10 marks)

1. In adolescence:
 - (a) Brain growth is faster than at any other stage in life
 - (b) Peak bone mass occurs before the growth spurt ends
 - (c) Girls physiologically increase their percentage body fat
 - (d) Type 2 obesity related diabetes remains a very rare problem
2. Supplementation with long-chain fatty acids EPA and DHA during pregnancy and infancy will likely benefit the following:
 - (a) Visual performance
 - (b) Motor performance
 - (c) Behaviour disorders
 - (d) Memory
3. Which one of these organs has the lowest energy expenditure per unit of body mass (kg)?
 - (a) Brain
 - (b) Kidney
 - (c) Liver
 - (d) Skeletal Muscle
4. It has been observed that a reduction in family size and improved sanitary condition have led to an increase in allergic conditions. The 'hygiene hypothesis' suggests that
 - (a) The frequent use of antiseptic wipes will reduce allergies
 - (b) Living on a working farm will help reducing the development of atopic diseases
 - (c) Particular care must be taken during breast feeding in high risk infants
 - (d) Living on a working farm will increase the development of atopic diseases
5. Bone mineral accretion
 - (a) Is rapid in the first 10 weeks of foetal life
 - (b) Is slow in infancy
 - (c) In childhood, adolescence and young adulthood predicts risk of fracture in old age
 - (d) Is particularly high after the menopause in women

True / False (10 marks)

Answer True or False

6. True or False: Infants born to mothers who are vegan may be at increased risk of Vitamin C deficiency.
7. True or False: Evaluating the safety of a food substance includes considering the daily intake of that substance by humans consuming the food.
8. True or False: The safety of pesticide chemicals, substances migrating from food packaging, and colour additives must be proven before their use in food.
9. True or False: Muscle triacylglycerol is the primary source of energy for short-term intense physical activities.
10. True or False: The pathogenesis of hyperglycemia in type 2 diabetes includes increased glucose production, impaired insulin secretion, and decreased glucose uptake.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the Mediterranean diet and analyze how its components may contribute to preventing age-related cognitive decline. Provide specific examples and evidence to support your analysis.
12. Discuss the role of plant sources in providing essential amino acids for human nutrition, and analyze the limitations some plants may have regarding specific amino acid content.

End of Paper